

Multimodal Image Understanding Pipeline

Hallucination Analysis Report

Author

Palak Bhardwaj

1. Hallucination Example

000000001532.jpg



Generated Output

This image shows a bunch of cars driving under a bridge.

The model identifies the key object as a truck and generates the explanation:
"The purpose of a truck is to provide food for the hungry."

2. Ground Truth vs Model Output

Ground Truth	Model Output
Cars	Truck
Highway Bridge	Truck
Traffic Scene	Food Distribution
Road Infrastructure	Humanitarian Activity

The generated explanation is not grounded in the visual content, demonstrating a clear hallucination phenomenon.

3. Hallucination Types

Visual Hallucination

Truck detected although cars are the dominant objects.

Semantic Hallucination

Food-distribution concepts introduced without visual evidence.

Reasoning Hallucination

Unsupported reasoning chain from traffic scene to humanitarian activity.

Reasoning Chain

Traffic Scene → Truck → Food Delivery → Feeding Hungry People

4. Root Causes and Future Work

Root Causes

- Dataset Bias
- Weak Visual Grounding
- Error Propagation

Future Work

- Hallucination Detection Module
- Confidence Scoring
- Visual Grounding Verification
- Multimodal Fact Checking
- Explainable Reasoning Chains

Conclusion

This case study highlights the importance of robust visual grounding and consistency verification in multimodal AI systems.